

ARCHIT SAXENA

Design Engineer | CFD Simulation Specialist | FEA / Structural Analysis

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PROFESSIONAL SUMMARY

Mechanical Design Engineer with 7+ years of experience delivering high-fidelity CFD simulations, FEA structural analysis, and end-to-end product development across industrial, energy, and advanced manufacturing sectors. Proficient in ANSYS Fluent, OpenFOAM, ANSYS Workbench, and leading CAD platforms. Proven ability to reduce prototyping cycles, improve product reliability, and translate simulation insights into manufacturable designs. Skilled in composite and additive manufacturing, DoE-based optimization, and cross-functional R&D collaboration.

CORE COMPETENCIES

CFD	ANSYS Fluent, OpenFOAM — multiphase flow, FSI, turbulence modeling, thermal/conjugate heat transfer, aerodynamic optimization	Manufacturing	Composite layup tooling, additive manufacturing (FDM/SLA), NDT, AS9100 quality compliance
FEA / CAE	ANSYS Workbench, SolidWorks Simulation — linear/non-linear static, dynamic, modal, fatigue, thermal analysis	Optimization	Design of Experiments (DoE), genetic algorithms, topology optimization, parametric studies
Meshing	Hypermesh, ANSA, ANSYS Meshing — structured/unstructured, hex-dominant, boundary-layer refinement	Programming	Python (automation, post-processing), MATLAB, Shell scripting, C/C++ basics
CAD	SolidWorks, CATIA, Siemens NX, AutoCAD — parametric modeling, assemblies, GD&T, DFM/DFA	Turbomachinery	Blade design, compressor/turbine flow analysis, ducted gas turbine CFD, performance mapping

PROFESSIONAL EXPERIENCE

R&D Engineer – Design & Simulation | Anadrone Systems Pvt Ltd Jan 2024 – Present

- ▶ Led full-cycle CFD simulation of ducted propulsion systems using ANSYS Fluent — modeled turbulent internal flows, optimized inlet geometry, and improved thrust-to-drag ratio by targeted airflow management.
- ▶ Executed structural FEA (static stress, fatigue, and modal analysis) on airframe assemblies including longerons, wing ribs, and fuselage panels; ensured compliance with operational load requirements.
- ▶ Performed aircraft sizing and propulsion trade studies — including engine selection, thrust-to-weight ratio, and fuel/energy capacity estimation for fixed-wing platforms.
- ▶ Designed composite manufacturing tooling and executed hands-on composite layup and FDM/SLA 3D printing for prototypes and production parts.
- ▶ Produced manufacturing drawings with full GD&T annotation; supported AS9100 quality compliance.

R&D Engineer – Product Design & Simulation | Arthlab Technologies Pvt Ltd Feb 2022 – May 2023

- ▶ Delivered FEA-driven structural and thermal analysis (steady-state and transient) for industrial automation products using ANSYS Workbench and SolidWorks Simulation.
- ▶ Conducted non-linear static and dynamic simulations for structural validation; prepared detailed technical reports with stress distributions, safety margins, and design recommendations for clients.
- ▶ Designed sheet metal enclosures and mechanical systems using DFM/DFA principles; created full manufacturing drawing packages with GD&T and BOM.
- ▶ Developed rapid prototypes using additive manufacturing (FDM); reduced concept-to-prototype lead time by iterating design-simulation cycles in-house.

Note: Jun 2023 – Jan 2024 — Independent CFD consulting and structured upskilling via Skill-Lync CFD Engineer Master Certificate Program.

Design & Simulation Engineer | Muvtons Castors Pvt Ltd Sep 2021 – Feb 2022

- ▶ Designed heavy-duty castor wheel assemblies; performed non-linear static and dynamic FEA to assess fatigue life, deformation, and safety margins under rated and overload conditions.
- ▶ Conducted experimental structural durability validation; correlated test results with simulation data and refined models accordingly.

- ▶ Generated technical reports with stress distribution contours, fatigue life curves, and safety margin summaries; created manufacturing drawings with GD&T and BOM.

Design & Simulation Engineer | Vyomnetra AI Pvt Ltd

Jun 2021 – Sep 2021

- ▶ Designed and analyzed structural assemblies for heavy-lift multi-rotor platforms; performed load analysis, static structural FEA, and CFD simulations to validate aerodynamic and structural performance.
- ▶ Created detailed engineering drawings with GD&T annotations and Bill of Materials for manufacture.

Design Engineering Intern | Hawaii-Adda Aerial Innovations Pvt Ltd

Nov 2020 – Feb 2021

- ▶ Supported CAD modeling, CFD/FEA simulation workflows for agricultural and urban UAV applications; contributed to component design validation and documentation.

Independent Design Engineering Consultant | Self-Employed

Oct 2016 – Mar 2019

- ▶ Delivered end-to-end mechanical design services (3D CAD, FEA, CFD, rapid prototyping) to diverse clients in product development, FMCG, and industrial equipment sectors.

KEY PROJECTS

EV Material Handling Trolley	Structural design and FEA-validated chassis for electric-assist trolley; topology optimization reduced weight by 18% while maintaining safety margins.
Automated Heavy-Duty Screw Jack	Mechanical design, kinematic analysis, and FEA of power screw and housing assembly for industrial lifting application.
Heavy-Payload Hexa-Copter	Full-frame structural design, load analysis, and CFD-validated propulsion layout for 25 kg payload multi-rotor.
High-Speed Aerial Target System	End-to-end design and CFD/FEA analysis of high-speed fixed-wing platform including ducted propulsion optimization.
CANSAT Mars Glider	Aerodynamic shaping, CFD simulation, and composite structure design for competition-class stratospheric glider.
Multipurpose QuadCopter	Structural and aerodynamic design, thrust-to-weight sizing, and FEA of frame and motor mounts.

EDUCATION

B.Tech – Mechanical Engineering | Rajasthan Technical University

2012 – 2016

PG Diploma – Computational Fluid Dynamics | 2020

CFD Engineer Master Certificate Program | Skill-Lync | Completed 2024

CERTIFICATIONS & LICENSES

DGCA Certified RPAS / Drone Pilot — License No. PC022400007GW | Valid 10 Years | 20+ Flight Hours

AS9100 Quality Management Standards — Aerospace quality and process compliance

NDT Techniques — Non-destructive testing methods for structural inspection